# The Evaluation of Convectively Coupled Waves in Limited-Length Datasets

**Chang Xu [1,2], Steven Sherwood [1,2], Martin Jucker [1,2], Clemente Lopez Bravo [1,2], Richard Jones [3]**

1 Climate Research Centre (CCRC), The University of New South Wales, Sydney NSW 2052, Australia
2 ARC Centre of Excellence for 21st Century Weather, Melbourne VIC 3800, Australia
3 The Met Office, Exeter, Devon, EX1 3PB, the United Kingdom

## Abstract

Convectively coupled waves (CCWs), including Kelvin waves and the MJO, play a key role in organising tropical convections. Kilometre-scale models enable explicit simulation of deep convection but are often limited to short outputs due to computational cost.

We assess the detectability of CCWs from datasets shorter than one year using FFT-based (WK-FFT) and wavelet-based (CFWT) spectral methods. Detectability is quantified using the Signal Ratio (SR) in zonal wavenumber–frequency space.

The CFWT consistently outperforms the WK-FFT approach, particularly for short datasets. Detection skill also depends on the variable used. Using the wavelet method, Kelvin waves can be robustly detected from≥90 days of OLR data, whereas the MJO can be detected from≥30 days of precipitation data. Applying this framework to six tropical kilometre-scale Unified Model simulations reveals a linked skill in simulating Kelvin waves and the MJO, suggesting common sensitivities in model design.

These results provide practical guidance for CCW detection in short model simulations and a benchmark for evaluating kilometre-scale tropical model performance.

## Methodology

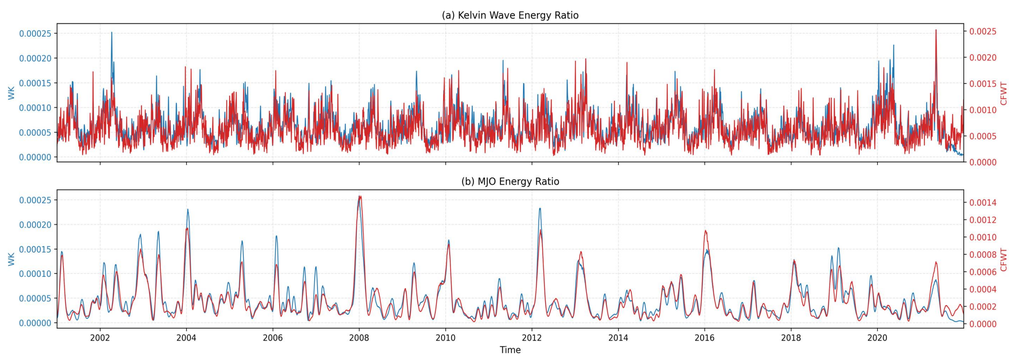
- Wheeler-Kiladis Filter (WK-FFT)
- The Combined-FFT Wavelet Transform (CFWT)
  - Zonal Wavenumber - Frequency
  - Temporal Changes in Waves
  - Energy Conservation

The Signal Ratio (SR)

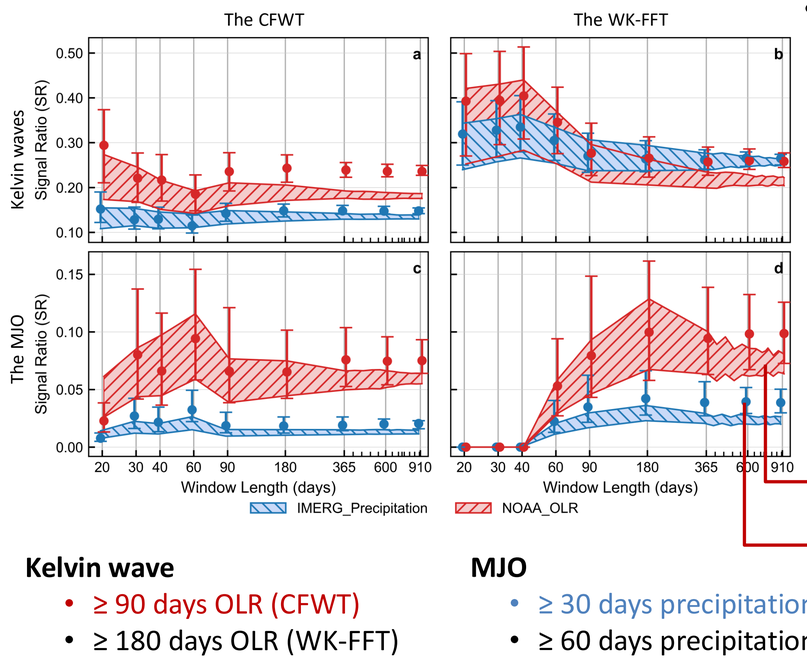
- Calculated from the domain after WK-FFT/ CFWT
- $SR_L = {}^{E_U}/_{E_T}$
- Detectability: observational SR above SR of red noise

## Results

### 1 CFWT is efficient in detecting CCWs in both long- and short-term dataset




- The CFWT gets similar signal intensity of both Kelvin waves and the MJO to that of WK1999 with using a long data record.




- The choice of variable is **as important as the detection method** for short datasets
  - **OLR** → better for **Kelvin wave** detection
  - **Precipitation** → better for **MJO** detection

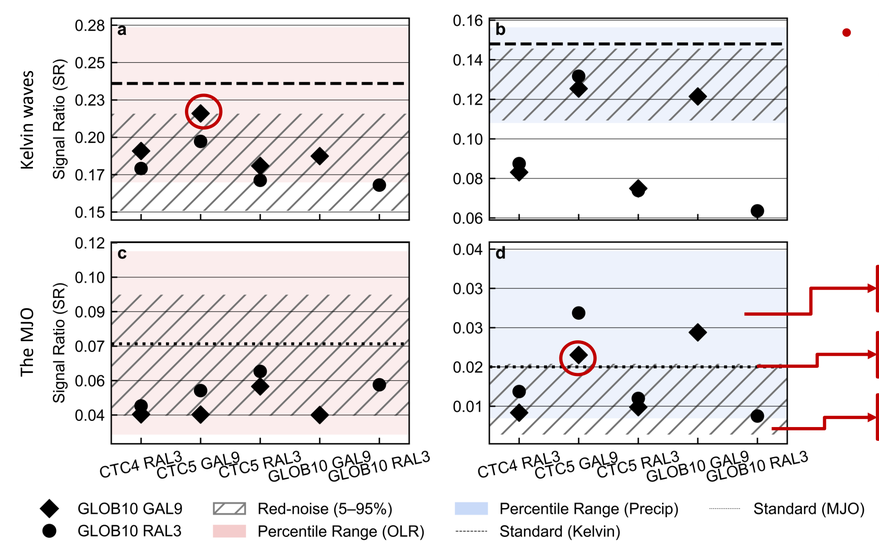
$[SR_{noise}]_{5-95\%}$

$[SR_{obs}]_{5-95\%}$

**Kelvin wave**
- ≥ 90 days OLR (CFWT)
- ≥ 180 days OLR (WK-FFT)

**MJO**
- ≥ 30 days precipitation (CFWT)
- ≥ 60 days precipitation (WK-FFT)

### 2 CCWs in Kilometre-scale Unified Model



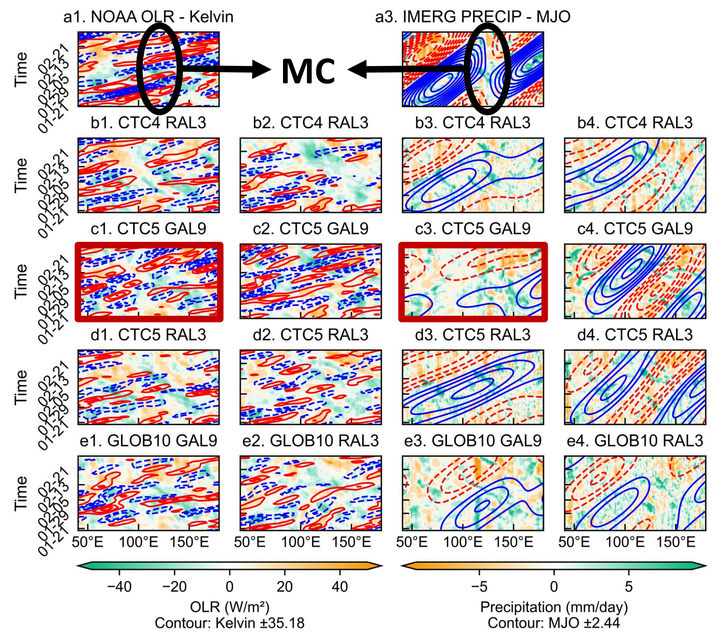

- Best-Performing Configuration: 5.5-km UM (CTC5 GAL9) + GLOB10 GAL9 forcing

$[SR_{obs}]_{5-95\%}$

SR Standard

$[SR_{noise}]_{5-95\%}$

- Higher resolution generally improves **Kelvin + MJO activity**
- Explicit convection (RAL3) remains **strongly influenced by boundary conditions**




- **The simulations of CCWs in MC are seen the significant biases**

## Conclusions

CFWT enables robust identification of Kelvin waves and the MJO in short-kilometre-scale simulations, with detection limits strongly dependent on the choice of variables. An SR-based framework provides a practical benchmark for screening CCW performance and guiding efficient model design under fixed computational resources

Email: chang.xu8@unsw.edu.au Phone: +61-414046641
Address: Climate Change Research Centre, Level 4, Mathews Building, The University of New South Wales, Sydney NSW 2052